Real-Life Problem Solving

Data Science Applications in the Real World — Case Studies in Action

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Detailed Response & Presentation Speaker Notes

Research base: 10 web & academic sources (NotebookLM). Every figure traces to a cited source.

PART 1 — Detailed Response

The Core Idea

Data science is not "math for its own sake." It is a repeatable way of turning messy, real-world data into a decision that saves money, time, or lives. The pattern is always the same: a painful business or human problem → data → a model → a measurable outcome. The case studies below prove it across finance, healthcare, and operations.

A. Finance & Banking — Catching Fraud and Forecasting Cash

1. Retail bank fraud overhaul (SG Analytics)

- **Problem:** Old rule-based system blocked genuine customers (payment holds, lost sales) without cutting fraud.
- **Method:** Semi-supervised ML with adaptive learning on transaction frequency, amount anomalies, geolocation.
- **Result: $6M** reduction in fraudulent transactions, plus fewer false positives.

2. Leading U.S. bank (DataWalk)

- **Problem:** Data scattered across business lines; no single view to catch complex fraud.
- **Method:** Agile data ingestion + knowledge-graph analytics.
- **Result: $40M+ saved annually**, stopping ~**$3.5M of fraud every month**.

3. HSBC × Google Cloud (Anti-Money-Laundering)

- **Problem:** AML alerts were mostly false alarms, wasting investigator time.
- **Method:** Dynamic Risk Assessment with ML.
- **Result: 2–4x more** true criminals caught while cutting alert volume by **~60%**.

4. Danske Bank

- **Problem:** Legacy rules engine drowning in false positives.

- **Method:** Deep-learning ensembles.
- **Result: ~60% fewer false positives**, **+50% true-fraud detection**.

5. Stripe Radar (used by LetsGetChecked)

- **Problem:** Block fraud without rejecting real buyers.
- **Method:** Network-level ML scoring every transaction.
- **Result: ~5x ROI**, scoring transactions in **<100 milliseconds**.

6. J.P. Morgan × Prysmian (cash-flow forecasting)

- **Problem:** Manual multi-entity cash forecasting.
- **Method:** AI cash-flow forecasting (Cash Flow Intelligence).
- **Result: ~50% less manual work**, **~$100K/year** saved.

7. Dr Pepper Snapple Group

- **Problem:** High cost and bottlenecks in financial-services transaction monitoring.
- **Method:** Predictive ML flagging outlier transactions in real time.
- **Result:** financial-services costs down **$2.5M**, with higher processing volume and quality.

> *"Banks turned fraud detection from a cost center into a $40M/year saving — by replacing rules with learning systems."*

B. Healthcare & Medicine — Predicting Disease Earlier

8. CardioGuard AI — KUST & PIMS (Pakistan)

- **Problem:** Manage cardiovascular disease better than the traditional Framingham Risk Score; 290 adults studied.
- **Method:** Random Forest + Logistic Regression in a real-time Streamlit clinical app on EHR data.
- **Result: 81.57% accuracy / 85.31% sensitivity**; operations: **ER visits −26.88%**, **admissions −12.50%**, **cardiac catheterizations −30.36%**, proactive surgeries **+33.33%**.

9. Breast cancer diagnosis (Wisconsin dataset)

- **Problem:** Earlier, more reliable breast-cancer detection.
- **Method:** Feedforward Neural Network vs Random Forest vs Decision Tree.
- **Result:** FNN won at **~97%** across accuracy/precision/recall/F1 (vs RF ~94%, DT ~94%).

10. Cerebral infarction prediction — Central China hospital (Chen et al.)

- **Problem:** Predict stroke / brain infarction onset from hospital records.
- **Method:** CNN on combined structured + unstructured (text) hospital data.
- **Result: 94.8% accuracy**, beating single-data-type models.

11. Cloud health forecasting (Sahoo et al.)

- **Method:** Probabilistic data collection + stochastic prediction model.
- **Result: ~98% prediction accuracy**, analysis time cut **90%**.

C. Healthcare Operations & Population Health — Named Hospitals

- **Massachusetts General Hospital** — queuing theory + simulation on patient flow/staffing → shorter waits, higher satisfaction, cost savings.
- **UCSF Health × GE Healthcare** — ML on ICU vitals + EHRs (GE Mural) to predict deterioration → lower ICU mortality, shorter stays.
- **Kaiser Permanente × IBM Watson Health** — predictive models on EHR + claims + social data to flag high-risk patients → fewer hospitalizations.
- **Cleveland Clinic** — NLP + ML on medication orders to cut errors → fewer adverse drug events.
- **BBVA AI Factory** — ML predicts loan delinquency early so the bank offers refinancing first → better debt recovery.

> *Aggregate: predictive analytics in hospital operations has driven up to **35% fewer readmissions**, **~30% lower mortality**, **30%+ shorter wait times**, and ER waits down **40%+**.*

D. How It's Actually Done — The Data Science Workflow

- **1. Gather & integrate** data from siloed sources (transactions, EHRs, clinical notes).
- **2. Clean & preprocess** — remove duplicates, handle missing values.

- **3. Feature engineering & scaling** — encode categories, reduce dimensions (PCA), standardize.
- **4. Train & validate** — split data (80/20 or chronological), test models (Random Forest, XGBoost, neural nets), k-fold cross-validation.
- **5. Tune & calibrate** — adjust thresholds to the business cost of a false negative vs false positive.
- **6. Deploy** — dashboards / clinical apps (Streamlit) or real-time pipelines scoring in milliseconds.

E. The Hard Part — Challenges, Limits, Ethics

- **Class imbalance:** fraud/disease is often <1% of data; naive "accuracy" looks great while catching nothing.
- **Cost of false positives:** blocked real customers, wasted investigator/clinician time.
- **Concept drift:** fraudsters adapt; yesterday's model goes stale.
- **Algorithmic bias:** AI under-diagnosed heart disease in women (male-dominated training data); worse on darker skin tones.
- **Black-box problem:** deep models can't explain themselves → breaks regulation (GDPR right to explanation) and clinician trust.
- **Privacy/security:** centralizing medical/financial records is a HIPAA/GDPR risk.

F. What Makes Projects Succeed (Lessons)

- **Use the right metric** — F1, Precision, Recall, AUPRC, not raw accuracy; cost-sensitive learning.
- **Explainable AI (XAI)** — SHAP / LIME open the black box and win trust + compliance.
- **Real-time pipelines** — scoring in <50–100ms, not nightly batches.
- **Fight imbalance** — SMOTE oversampling + ensembles (RF, XGBoost).
- **Federated learning** — train across hospitals/banks without moving raw data (privacy-safe).
- **Human-in-the-loop** — AI augments investigators and clinicians, doesn't replace them.

PART 2 — Presentation Speaker Notes

Opening hook (say this)

> *"Every time your bank blocks a fraudulent charge, a hospital catches a disease early, or Netflix knows what you'll watch next — that's data science solving a real problem. Today I'll show you it in action, with the actual numbers."*

Per-section notes

- **Why this matters:** Frame DS as problem → data → model → measurable outcome. Don't define algorithms; define impact.
- **Finance:** Lead with the $40M DataWalk number, it lands hardest. Contrast old rules vs learning systems. Point: same data, smarter method, 60% fewer false alarms.
- **Healthcare:** Use CardioGuard AI as your hero story (local/Pakistan, relatable, has operational numbers, not just accuracy). 97% breast-cancer model = "earlier detection saves lives."
- **Operations:** Named hospitals (MGH, UCSF, Kaiser, Cleveland Clinic) show DS isn't just diagnosis — it runs the hospital. Use the aggregate: 35% fewer readmissions.
- **How it works:** Walk the 6-step pipeline once, plainly. This is the "in action" part — show it's a craft, not magic.
- **Challenges/Ethics:** Be honest. The mark of maturity: "high accuracy can still be useless." Bias example (women under-diagnosed) gets the room's attention.
- **Success factors:** End on what works: right metrics, explainable AI, human-in-the-loop. This is your "so what."

Closing line

> *"Data science wins when it's measured in dollars saved, wait times cut, and lives extended — not in model accuracy alone."*

Anticipated Q&A

"Will AI replace doctors/analysts?" → No — every successful case is human-in-the-loop; AI flags, humans decide.

"Isn't accuracy the goal?" → No, with rare events accuracy is misleading; precision/recall/AUPRC matter.

"Biggest risk?" → Bias and the black-box problem; mitigated by XAI and diverse data.

Honesty note: For MGH, UCSF, Kaiser, Cleveland Clinic, and BBVA the sources describe measurable improvements but do not publish exact figures — present these as qualitative wins, not invented numbers.